Exercise Solution

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3.2 The voltage level of both CMOS 74HCT and TTL 74ALS is summarized in Table X3-1. Suppose that TTL 74ALS drives CMOS 74HCT. Please calculate the DC noise margin, (1) when the output voltage of TTL 74ALS is high, and (2) when the output voltage of TTL 74ALS is low.

Table X3-1 Voltage Level of CMOS 74HCT and TTL 74ALS

Logic Family	Voltage Level			
	V_{OHmin}	V_{IHmin}	V_{ILmax}	V_{OLmax}
74HCT	3.84 V	2.0 V	0.8 V	0.33 V
74ALS	2.7 V	2.0 V	0.8 V	0.5 V

Suppose that TTL 74ALS drives the CMOS 74HCT. When the output voltage of TTL 74ALS is high, the DC noise margin is calculated as:

$$N_H = V_{OHmin} - V_{IHmin} = 2.7\text{V} - 2.0\text{V} = 0.7\text{V}$$

When the output voltage of TTL 74ALS is low, the DC noise margin is calculated as:

$$N_L = V_{ILmax} - V_{OLmax} = 0.8\text{V} - 0.5\text{V} = 0.3\text{V}$$

The DC noise margin "N_H" and "N_L" are positive in this case. Therefore, when TTL 74ALS drives CMOS 74HCT, the interface is compatible.

3.3 The current level of both CMOS 74HCT and TTL 74AS is summarized in Table X3-2. Suppose that CMOS 74HCT drives TTL 74AS. Please calculate (1) the overall fanout of CMOS 74HCT, and (2) the residual driving capacity of CMOS 74HCT.

Table X3-2 Current Level of CMOS 74HCT and TTL 74AS

Logic Family	Current Level			
	I_{OHmax}	I_{OLmax}	I_{IHmax}	I_{ILmax}
74HCT	4mA	4mA	1μA	1μA
74AS	2mA	20mA	0.02mA	0.5mA

(1) The low-state fanout of CMOS 74HCT is calculated as:

$$F_L = \frac{I_{OLmax}}{I_{ILmax}} = \frac{4mA}{0.5mA} = 8$$

The high-state fanout of CMOS 74HCT is calculated as:

$$F_H = \frac{I_{OHmax}}{I_{IHmax}} = \frac{4mA}{0.02mA} = 200$$

The overall fanout of CMOS 74HCT is calculated as:

$$F_O = \min(F_L, F_H) = \min(200, 8) = 8$$

Therefore, the overall fanout of CMOS 74HCT is “8” TTL 74AS logic gates.

(2) The residual driving capacity of CMOS 74HCT is calculated as:

$$\begin{aligned} R_C &= I_{OHmax} - F_O \times I_{IHmax} \\ &= 4mA - 8 \times 0.02mA \\ &= 4mA - 0.16mA \\ &= 3.84mA \end{aligned}$$

Therefore, the residual driving capacity of CMOS 74HCT is “3.84mA”.

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3.7 Figure X3-1(1) Write the voltage table and the logic function. (2) Draw the logic circuit diagram of the logic function.

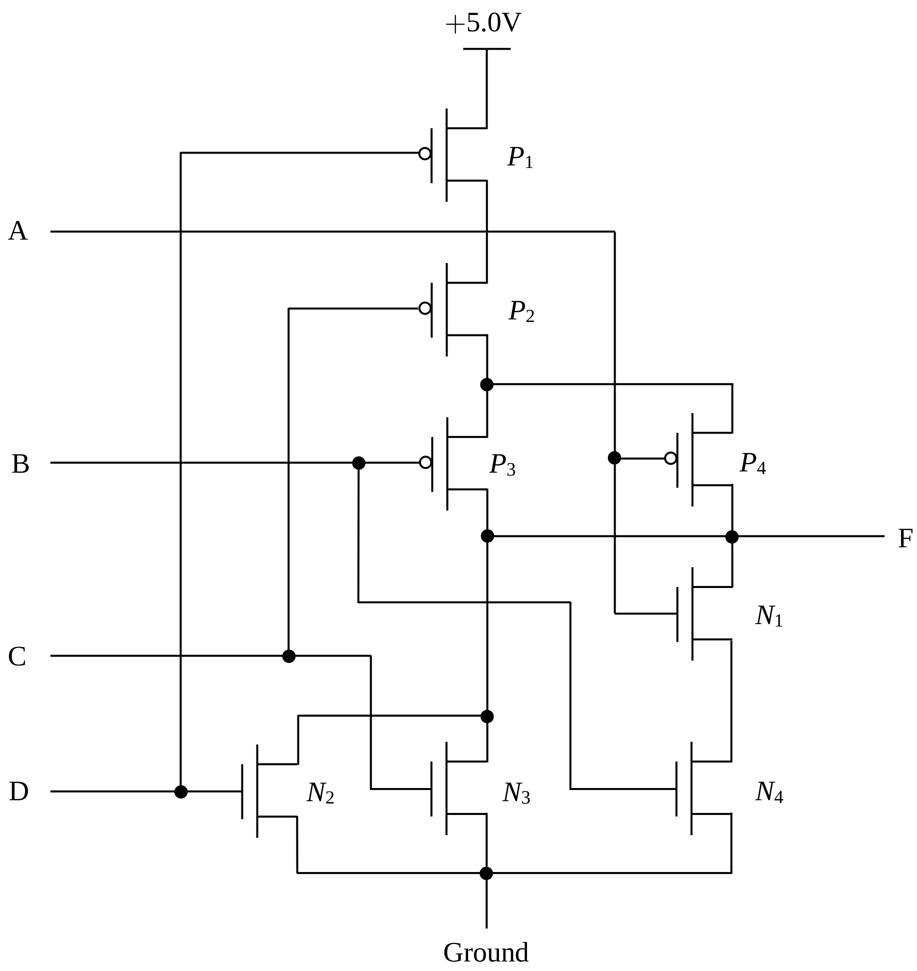
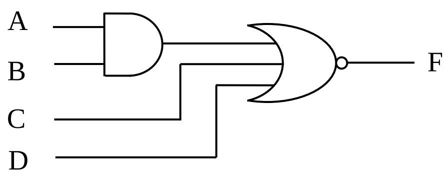


Figure X3-1 Example CMOS Digital Circuit of Exercise 3.4

$$F = f(A, B, C, D) = \overline{A \cdot B + C + D}$$

Voltage Table of CMOS Digital Circuit of Exercise 3.4

Inputs				PMOS				NMOS				Output
A	B	C	D	P_1	P_2	P_3	P_4	N_1	N_2	N_3	N_4	F
V_L	V_L	V_L	V_L	ON	ON	ON	ON	OFF	OFF	OFF	OFF	V_H
V_L	V_L	V_L	V_H	OFF	ON	ON	ON	OFF	ON	OFF	OFF	V_L
V_L	V_L	V_H	V_L	ON	OFF	ON	ON	OFF	OFF	ON	OFF	V_L
V_L	V_L	V_H	V_H	OFF	OFF	ON	ON	OFF	ON	ON	OFF	V_L
V_L	V_H	V_L	V_L	ON	ON	OFF	ON	OFF	OFF	OFF	ON	V_H
V_L	V_H	V_L	V_H	OFF	ON	OFF	ON	OFF	ON	OFF	ON	V_L
V_L	V_H	V_H	V_L	ON	OFF	OFF	ON	OFF	OFF	ON	ON	V_L
V_L	V_H	V_H	V_H	OFF	OFF	OFF	ON	OFF	ON	ON	ON	V_L
V_H	V_L	V_L	V_L	ON	ON	ON	OFF	ON	OFF	OFF	OFF	V_H
V_H	V_L	V_L	V_H	OFF	ON	ON	OFF	ON	ON	OFF	OFF	V_L
V_H	V_L	V_H	V_L	ON	OFF	ON	OFF	ON	OFF	ON	OFF	V_L
V_H	V_L	V_H	V_H	OFF	OFF	ON	OFF	ON	ON	ON	OFF	V_L
V_H	V_H	V_L	V_L	ON	ON	OFF	OFF	ON	OFF	OFF	ON	V_L
V_H	V_H	V_L	V_H	OFF	ON	OFF	OFF	ON	ON	OFF	ON	V_L
V_H	V_H	V_H	V_L	ON	OFF	OFF	OFF	ON	OFF	ON	ON	V_L
V_H	V_H	V_H	V_H	OFF	OFF	OFF	OFF	ON	ON	ON	ON	V_L



Logic Circuit Diagram of Exercise 3.4